10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 21 — Organic Chemistry

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 21.1

1. What is the difference between cyclohexane and benzene?

Cyclohexane	Benzene
Its molecular formula is C_6H_{12}.	Its molecular formula is C_6H_6.
It is a saturated alicyclic compound; all the carbon–carbon bonds in its ring are single bonds.	It is an aromatic compound; its six-carbon ring has three alternate double and single bonds, whose electrons are spread over the whole ring.
Each carbon atom is bonded to two hydrogen atoms, so it has no free valency to add more atoms.	Each carbon atom is bonded to only one hydrogen atom, so the ring is unsaturated.
Like alkanes, it does not undergo addition reactions and does not decolourise bromine water.	Being aromatic, it is unsaturated and gives substitution reactions rather than simple addition reactions.

2. Give two examples of branched chain compounds.

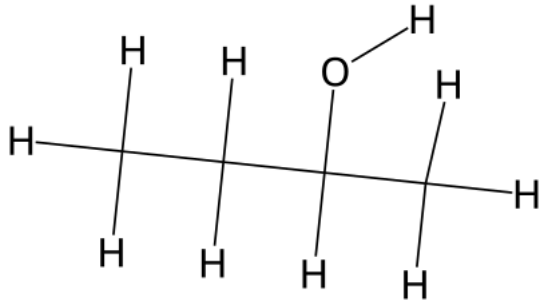
2-Methylpropane (isobutane), $CH_3–CH(CH_3)–CH_3$, and **2-methylbutane** (isopentane), $CH_3–CH(CH_3)–CH_2–CH_3$, are two examples of branched chain compounds. In these compounds a methyl group, $–CH_3$, is attached to a carbon atom of the main chain instead of forming a straight chain.

Quick Check 21.2

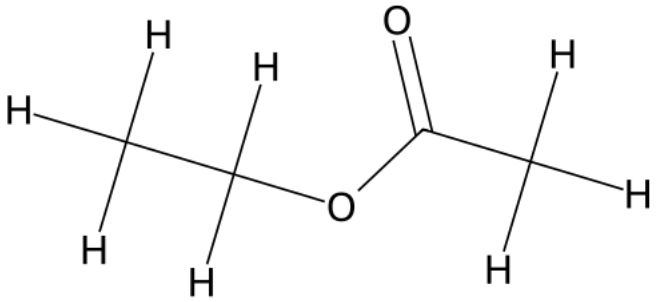
1. Write down the displayed and condensed formulae of butan-2-ol, ethyl ethanoate, 2-methylpropane, but-2-ene and pentane.

The condensed structural formulae are written in one line and the carbon–hydrogen bonds are omitted, while the displayed formula shows every atom and every bond by lines.

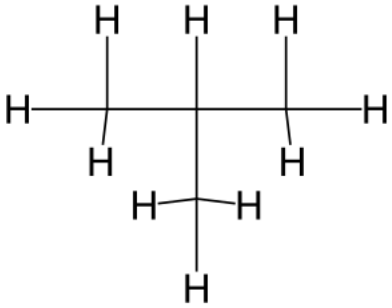
Butan-2-ol: $CH_3–CH(OH)–CH_2–CH_3$ (the –OH group is on carbon-2)



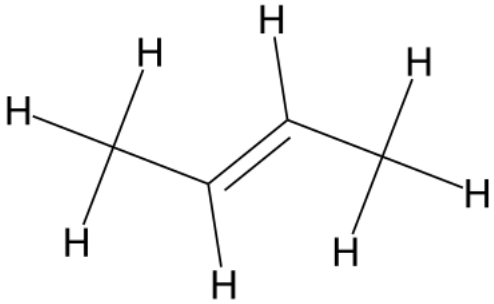
Ethyl ethanoate: $CH_3–C(=O)–O–CH_2–CH_3$ (an ester)



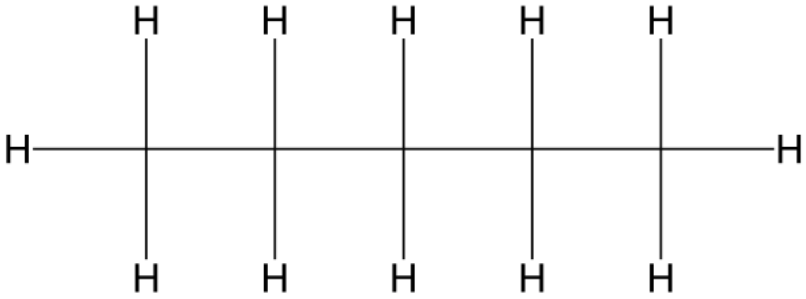
2-Methylpropane: $CH_3–CH(CH_3)–CH_3$ (a branched chain alkane)



But-2-ene: $CH_3–CH=CH–CH_3$ (an alkene with the double bond in the middle)



Pentane: $CH_3–CH_2–CH_2–CH_2–CH_3$ (a straight chain alkane)



Quick Check 21.4

1. How are phenols different from alcohols?

Phenols	Alcohols
In phenols the –OH group is attached directly to the carbon atom of an aromatic ring, such as the benzene ring.	In alcohols the –OH group is attached to a saturated carbon atom of an aliphatic chain.
Their general formula is Ar–OH, and the simplest member is phenol, C_6H_5OH.	Their general formula is R–OH, and a simple member is ethanol, CH_3CH_2OH.
The –OH group of a phenol is acidic in nature because the ring withdraws electrons, so phenols dissolve in alkalis.	Alcohols are neutral and do not react with alkalis.
Phenol is more reactive than alcohols and gives substitution reactions on the ring.	Alcohols give typical reactions of the hydroxyl group, such as esterification and oxidation.

2. What is common between the aldehydic and ketonic functional groups?

Both the aldehydic and the ketonic functional groups contain the carbonyl group, C=O. In an aldehyde group, written –CHO, the carbonyl carbon is bonded to at least one hydrogen atom, as in methanol (formaldehyde) HCHO and ethanal (acetaldehyde) CH_3CHO. In a ketone group the carbonyl carbon is bonded to two carbon groups, as in propanone CH_3COCH_3. Thus both groups have the same carbon–oxygen double bond, but they differ in the groups attached to the carbonyl carbon, and both are formed by the oxidation of alcohols.

Quick Check 21.5

1. Name the following compounds according to the IUPAC system of nomenclature: (i) CH_3-CH_2-CH_2-CH_2-CH_3 (ii) CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3

(i) The longest chain contains five carbon atoms, so the root is ‘pent’ and all the bonds are single, so the name of this straight chain alkane is **pentane**.

(ii) The longest chain contains seven carbon atoms, so the root is ‘hept’ and the compound is a saturated hydrocarbon; the IUPAC name is **heptane**, $CH_3–(CH_2)_5–CH_3$.

Quick Check 21.6

1. Condensed structural formulae of some organic compounds are given below. Write down the names of their families.

Compound	Family
(i) CH_3–CH(OH)–CH_3	Alcohol (propan-2-ol)
(ii) CH_3–CH_2–C(=O)–CH_3	Ketone (propanone)
(iii) CH_3–C(CH_3)=CH–CH_3	Alkene (an unsaturated hydrocarbon)
(iv) CH_3–CH_2–CH_2–COOH	Carboxylic acid (butanoic acid)
(v) C_3H_6	Alkene — with the general formula C_nH_{2n} it is propene, CH_2=CH–CH_3. (The same formula can also represent the ring compound cyclopropane.)
(vi) C_5H_{12}	Alkane — it fits the general formula C_nH_{2n+2}, so it is pentane.
(vii) H–C(=O)–H	Aldehyde (methanal or formaldehyde)
(viii) CH_3–CH_2–C(=O)–O–CH_2–CH_3	Ester (ethyl propanoate)

Exercise (Solved)

A. Multiple Choice Questions

1. Choose the correct IUPAC name for the compound HC(=CH_2)–CH_2–CH_3, that is CH_2=CH–CH_2–CH_3:

(a) But-3-ene (b) But-1-ene
(c) But-2-ene (d) Butene

Answer: (b) — The chain has four carbon atoms and the double bond starts at carbon-1, so the name is but-1-ene.

2. Select a branched chain compound.

(a) CH_2=CH–CH_2–CH_3 (b) CH_2=C(CH_3)–CH_2–CH_3
(c) CH_3–CH=CH–CH_3 (d) CH_3–CH=CH–CH_2–CH_3

Answer: (b) — In option (b) a methyl group, –CH_3, is attached to carbon-2 of the chain, so the chain is branched; the others are straight chains.

3. Which is the displayed formula of ethanol?

(a) C_2H_5OH
(b) $CH_3–CH_2–OH$
(c) The structure in which the carbon–hydrogen bonds are not shown
(d) The structure showing every atom and every bond, including the O–H bond

Answer: (d) — A displayed formula shows all the atoms and all the bonds of the molecule, including the O–H bond; (a) is a molecular formula and (b) is a condensed formula.

4. Identify the aldehydic functional group among the following:

(a) –C(=O)OH
(b) –OH
(c) –C(=O)–
(d) –C(=O)H

Answer: (d) — The aldehydic group is –CHO, in which the carbonyl carbon carries a hydrogen atom; (a) is the carboxyl group, (b) the hydroxyl group and (c) the ketonic group.

5. Select the general formula of the alkyne family.

(a) C_nH_{2n+1}
(b) C_nH_{2n}
(c) C_nH_{2n-2}
(d) C_nH_{2n-1}

Answer: (c) — Alkynes contain a carbon–carbon triple bond, so their general formula is C_nH_{2n-2}.

6. How many structural isomers will be there for a saturated hydrocarbon having five carbon atoms?

(a) 2
(b) 3
(c) 4
(d) 5

Answer: (b) — The three isomers of C_5H_{12} are pentane, 2-methylbutane and 2,2-dimethylpropane.

7. Which ester among the following will be called ethyl propanoate?

(a) $CH_3–C(=O)–OCH_3$
(b) $CH_3–CH_2–C(=O)–OCH_3$
(c) $CH_3–C(=O)–OC_2H_5$
(d) $CH_3–CH_2–C(=O)–OC_2H_5$

Answer: (d) — Ethyl propanoate is made from propanoic acid and ethanol, so it has a three-carbon acid part ($CH_3CH_2CO–$) and an ethyl group ($–OC_2H_5$) on the oxygen.

8. Which functional group is present in alkenes?

(a) No functional group
(b) Carbon–carbon triple bond
(c) Carbon–carbon double bond
(d) Carbon–carbon single bond

Answer: (c) — Alkenes are unsaturated hydrocarbons containing at least one carbon–carbon double bond, C=C, which is their functional group.

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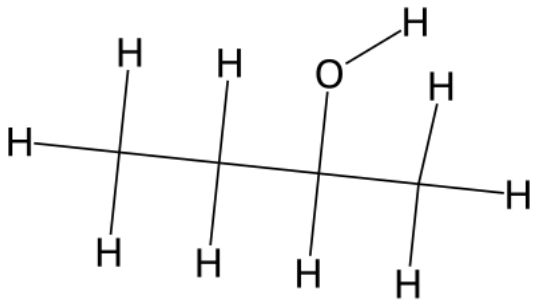
B. Short Answer Questions

21.1 What is the difference between saturated and unsaturated hydrocarbons?

Saturated hydrocarbons	Unsaturated hydrocarbons
All the carbon–carbon bonds in their molecules are single bonds.	Their molecules contain at least one carbon–carbon double bond or triple bond.
The four valencies of every carbon atom are satisfied by single bonds, so no more atoms can be added.	They have one or more pairs of carbon atoms joined by a double or a triple bond, so more atoms can be added across these bonds.
Their general formula is C_nH_{2n+2}, for example ethane CH_3–CH_3.	Alkenes have the formula C_nH_{2n}, for example ethene CH_2=CH_2; alkynes have C_nH_{2n-2}, for example ethyne HC≡CH.
They do not decolourise bromine water in the dark and give substitution reactions.	They decolourise bromine water and cold dilute $KMnO_4$ solution, and give addition reactions.

21.2 Write down the structural formula of butan-2-ol.

Butan-2-ol has four carbon atoms in a chain and the hydroxyl group, –OH, is attached to the second carbon atom. Its structural (condensed) formula is CH_3–CH(OH)–CH_2–CH_3, and its structural formula is written below; its molecular formula is C_4H_9OH.



21.3 Why is ethylamine, $CH_3CH_2NH_2$, called an organic compound?

Ethylamine is called an organic compound because its molecule is built on a framework of carbon atoms which are covalently bonded to hydrogen atoms. It is a derivative of ethane, CH_3CH_3, in which one hydrogen atom has been replaced by an amino group, $–NH_2$. Since organic compounds are the carbon compounds containing carbon–hydrogen bonds, and their derivatives, ethylamine is an organic compound; it belongs to the family of amines.

21.4 Write the formula of ethyl methanoate.

Ethyl methanoate is an ester made from methanoic acid (formic acid) and ethanol, so its formula is H–C(=O)–O–CH_2–CH_3, which is also written as $HCOOC_2H_5$. Its molecular formula is $C_3H_6O_2$.

21.5 Define structural isomerism.

Structural isomerism is the existence of two or more compounds which have the same molecular formula but different structural formulae, that is, their atoms are joined together in different ways. Such compounds are called structural isomers and they have different physical properties and sometimes different chemical properties. For example, n-butane $CH_3CH_2CH_2CH_3$ and 2-methylpropane $CH_3CH(CH_3)CH_3$ are structural isomers, because both have the molecular formula C_4H_{10}.

21.6 Give the IUPAC name of CH_3-CH_2-CH_2-CH_3.

The compound contains four carbon atoms in an unbranched chain and all the carbon–carbon bonds are single bonds, so the root is ‘but’ and the suffix is ‘ane’. Its IUPAC name is therefore **butane** (common name n-butane), and its molecular formula is C_4H_{10}.

21.7 What is common in the carboxyl, amide and ester functional groups?

All three groups contain the carbonyl group, C=O, in which a carbon atom is joined to an oxygen atom by a double bond. Their groups are –C(=O)OH (carboxyl), $–C(=O)NH_2$ (amide) and –C(=O)OR (ester), where R is an alkyl group. Because of this common carbonyl group, all three families show some similar reactions, and esters and amides are both derived from carboxylic acids.

C. Constructed Response Questions

21.1 Why do members of a homologous series have similar chemical properties?

The members of a homologous series all contain the same functional group, and the chemical properties of an organic compound are mainly determined by its functional group. For example, all alcohols contain the –OH group, so they all react with sodium to give hydrogen, and they all form esters with carboxylic acids. The members differ only in the length of their carbon chain, which affects their physical properties such as boiling point, but not their chemical behaviour. That is why the members of a homologous series have similar chemical properties.

21.2 How do the properties of 2-methylpropane and n-butane differ?

2-Methylpropane and n-butane are structural isomers with the same molecular formula C_4H_{10}, but they differ in structure and in physical properties.

2-Methylpropane (isobutane)	n-Butane
It has a branched structure: $CH_3CH(CH_3)CH_3$.	It has a straight chain: $CH_3CH_2CH_2CH_3$.
The molecule is more compact, so the molecules have less surface contact and weaker intermolecular forces.	The molecules are longer and have greater surface contact, so the intermolecular forces are stronger.
Because of weaker attraction its boiling point is lower (about −12 °C).	Because of stronger attraction its boiling point is higher (about −0.5 °C).

2-Methylpropane (isobutane)	n-Butane
Both are alkanes, so their chemical properties are similar; both burn to give carbon dioxide and water.	Both are alkanes, so their chemical properties are similar; both burn to give carbon dioxide and water.

21.3 Write the structural formulae of the isomers of the five-carbon saturated hydrocarbon.

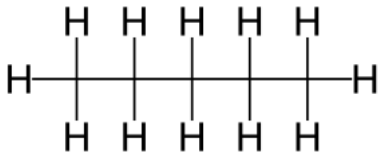
A saturated hydrocarbon with five carbon atoms has the molecular formula C_5H_{12} and it has three structural isomers.

(i) Pentane (straight chain): $CH_3–CH_2–CH_2–CH_2–CH_3$

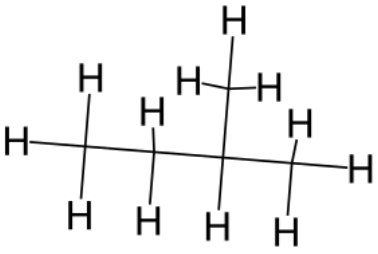
(ii) 2-Methylbutane (one branch): $CH_3–CH(CH_3)–CH_2–CH_3$

(iii) 2,2-Dimethylpropane (two branches on the same carbon): $CH_3–C(CH_3)_2–CH_3$

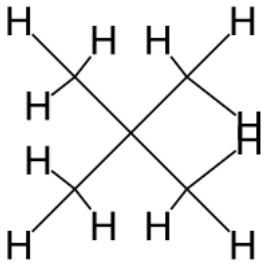
Pentane



2-Methylbutane



2,2-Dimethylpropane



All three compounds have the same molecular formula, C_5H_{12}, but their chains are arranged differently, so they are structural isomers of each other.

21.4 Write the structural formulae of the isomers of butene.

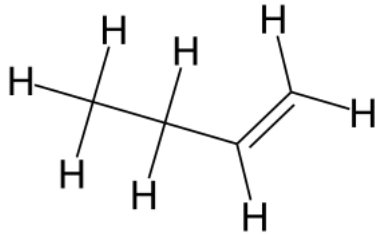
Butene has the molecular formula C_4H_8, and its isomers differ in the position of the double bond or in the arrangement of the chain.

(i) But-1-ene (double bond at carbon-1): $CH_2=CH–CH_2–CH_3$

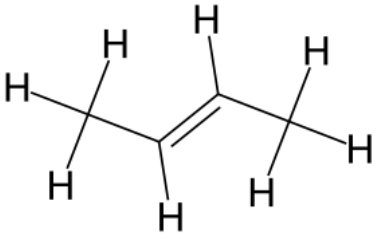
(ii) But-2-ene (double bond at carbon-2): $CH_3–CH=CH–CH_3$

(iii) 2-Methylpropene (branched chain): $CH_2=C(CH_3)_2$

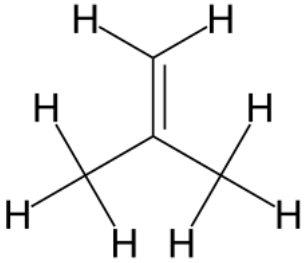
But-1-ene



But-2-ene



2-Methylpropene



The textbook shows the two isomers of butene, but-1-ene and but-2-ene, which differ in the position of the double bond; the branched isomer 2-methylpropene is also an isomer of butene.

21.5 Explain the difference among the functional groups in aldehydes, ketones and carboxylic acids.

Aldehyde	Ketone	Carboxylic acid
–C(=O)H, that is –CHO	–C(=O)–, that is a carbonyl group between two carbon atoms	–C(=O)OH, that is –COOH
The carbonyl carbon has at least one hydrogen atom attached to it.	The carbonyl carbon has two carbon groups attached to it.	The carbonyl carbon has an –OH group attached to it.
Example: CH_3CHO (ethanal)	Example: CH_3COCH_3 (propanone)	Example: CH_3COOH (ethanoic acid)

In an aldehyde the carbonyl group is at the end of the chain, in a ketone it is within the chain, and in a carboxylic acid the carbonyl group is combined with a hydroxyl group, which makes the compound acidic.

D. Descriptive Questions

21.1 What is an organic compound? How are organic compounds classified by their carbon chains?

An organic compound is a chemical compound which is primarily composed of carbon atoms that are covalently bonded to each other or to other elements. Apart from carbon, most organic compounds also contain hydrogen, and they may also contain oxygen, nitrogen, sulphur and halogens. For historical and conventional reasons, a few carbon compounds such as carbon oxides, carbonates and hydrogen carbonates are studied as inorganic compounds. Organic chemistry is the branch of chemistry in which we study the compounds of carbon and hydrogen (hydrocarbons) and their derivatives.

By their carbon chains, organic compounds are classified into two main classes.

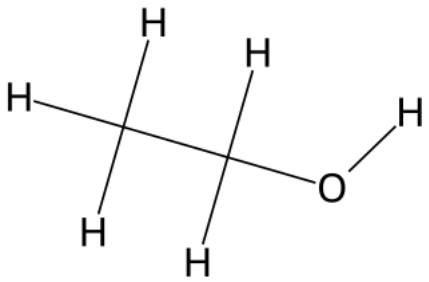
(i) Open chain or acyclic compounds: these compounds contain an open chain of carbon atoms, which may be straight (non-branched) or branched. For example, butane $CH_3CH_2CH_2CH_3$ is a straight chain compound, while 2-methylpropane $CH_3CH(CH_3)CH_3$ is a branched chain compound. Open chain compounds, together with certain cyclic compounds, are also called aliphatic compounds.

(ii) Closed chain or cyclic compounds: these compounds contain one or more closed chains (rings) of atoms. If the ring contains only carbon atoms, the compound is called carbocyclic, and if the ring also contains other atoms such as nitrogen or oxygen, it is called heterocyclic. Carbocyclic compounds are further divided into alicyclic compounds, such as cyclohexane, which contain a ring of three or more carbon atoms and resemble aliphatic compounds, and aromatic compounds, such as benzene, which contain a benzene ring with three alternate double and single bonds.

21.2 Explain structural formula with examples.

A structural formula of a compound shows the arrangement of the atoms and the bonds present in its molecule; it gives enough information to understand the structure of the molecule clearly. While writing it, most of the single covalent bonds are usually omitted and only important bonds such as double and triple bonds are shown.

Displayed formula: it shows the detailed structure of the molecule, that is, the relative positions of all the atoms and the number and type of bonds between them, and every bond is represented by a line. For example, the displayed formula of ethanol shows every C–H, C–C, C–O and O–H bond.



Condensed structural formula: it still uses lines between the bonded atoms but omits the carbon–hydrogen bonds; for example, the condensed formula of ethanol is CH_3–CH_2–OH. In the same way ethene is written CH_2=CH_2, which shows its double bond, and ethyl ethanoate is written CH_3–C(=O)–O–CH_2–CH_3, which shows its ester group.

21.3 Write five characteristics of a homologous series.

(i) All the members of a homologous series contain the same functional group.

(ii) They all share the same general formula, for example C_nH_{2n+2} for alkanes and C_nH_{2n} for alkenes.

(iii) Any two successive members differ by one –CH_2– unit, which is a difference of 14 in relative molecular mass.

(iv) All the members have similar chemical properties, because their chemical behaviour is determined by the same functional group.

(v) Their physical properties change gradually as the size of the molecule increases, for example the boiling point and the density increase with the increase in molecular mass, while the solubility in water usually decreases.

21.4 Give the IUPAC rules for naming alcohols.

(i) Select the longest continuous chain of carbon atoms which contains the carbon atom bonded to the hydroxyl group, and take it as the parent hydrocarbon; the ending ‘e’ of the alkane name is replaced by ‘ol’.

(ii) Number the carbon atoms of this chain from the end which gives the carbon atom attached to the –OH group the lowest possible number.

(iii) Indicate the position of the hydroxyl group by the number of its carbon atom, placed before –ol; if there are side groups, name and number them also and write them as prefixes.

By these rules, $CH_3CH_2CH_2OH$ is propan-1-ol, $CH_3CH(OH)CH_3$ is propan-2-ol, $CH_3CH_2CH_2CH_2OH$ is butan-1-ol and $CH_3CH(OH)CH_2CH_3$ is butan-2-ol.

21.5 Name the following compounds.

(a) $CH_3CH_2CH_2OH$ — **propan-1-ol** (a three carbon alcohol with the –OH on carbon-1).

(b) CH_3CH_2CHO — **propanal** (a three carbon aldehyde; the –CHO group is always at the end of the chain, so no number is needed).

(c) $CH_3CH_2CH_2COOH$ — **butanoic acid** (a four carbon carboxylic acid; the carboxyl carbon is always counted as carbon-1).

(d) $CH_3CH_2C(=O)OCH_3$ — **methyl propanoate** (an ester made from propanoic acid and methanol; the alkyl group of the alcohol is named first).

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